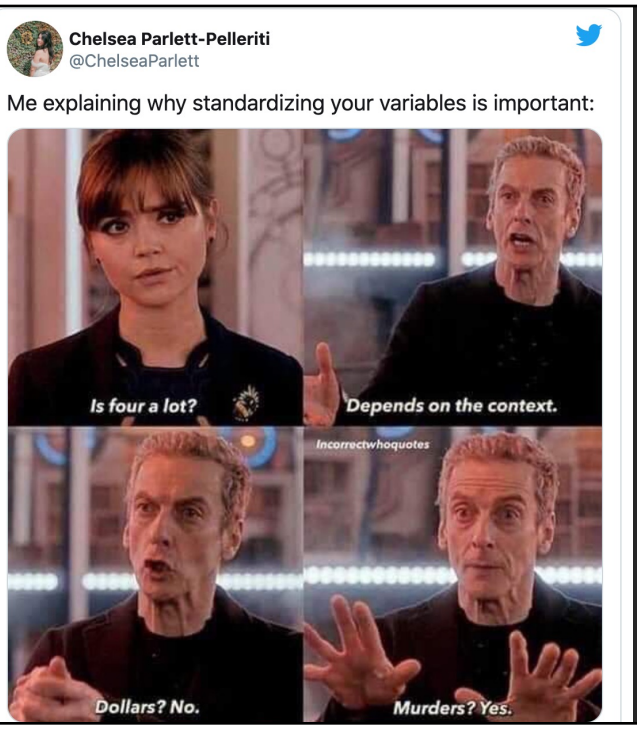
Communicative Visualization
SI649/EECS548

October 9, 2023

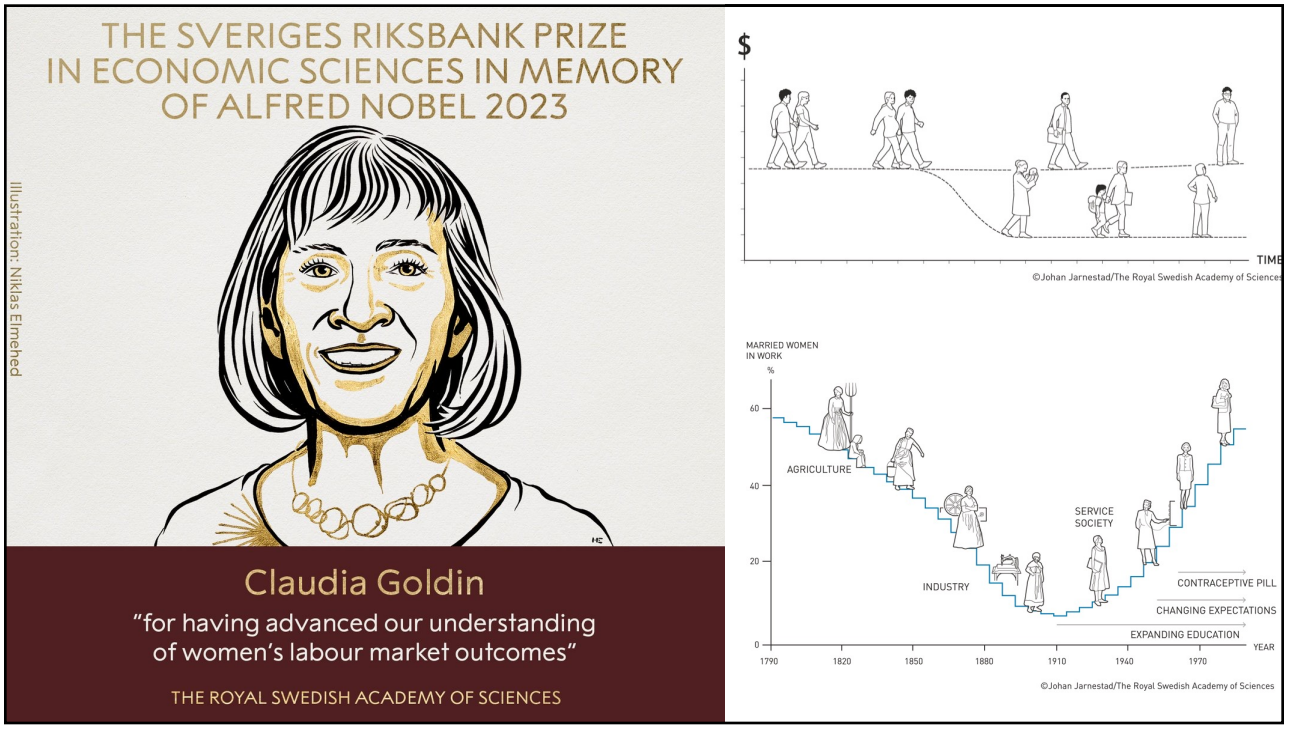
Some material adapted from Kay, Hullman & Agrawala

1

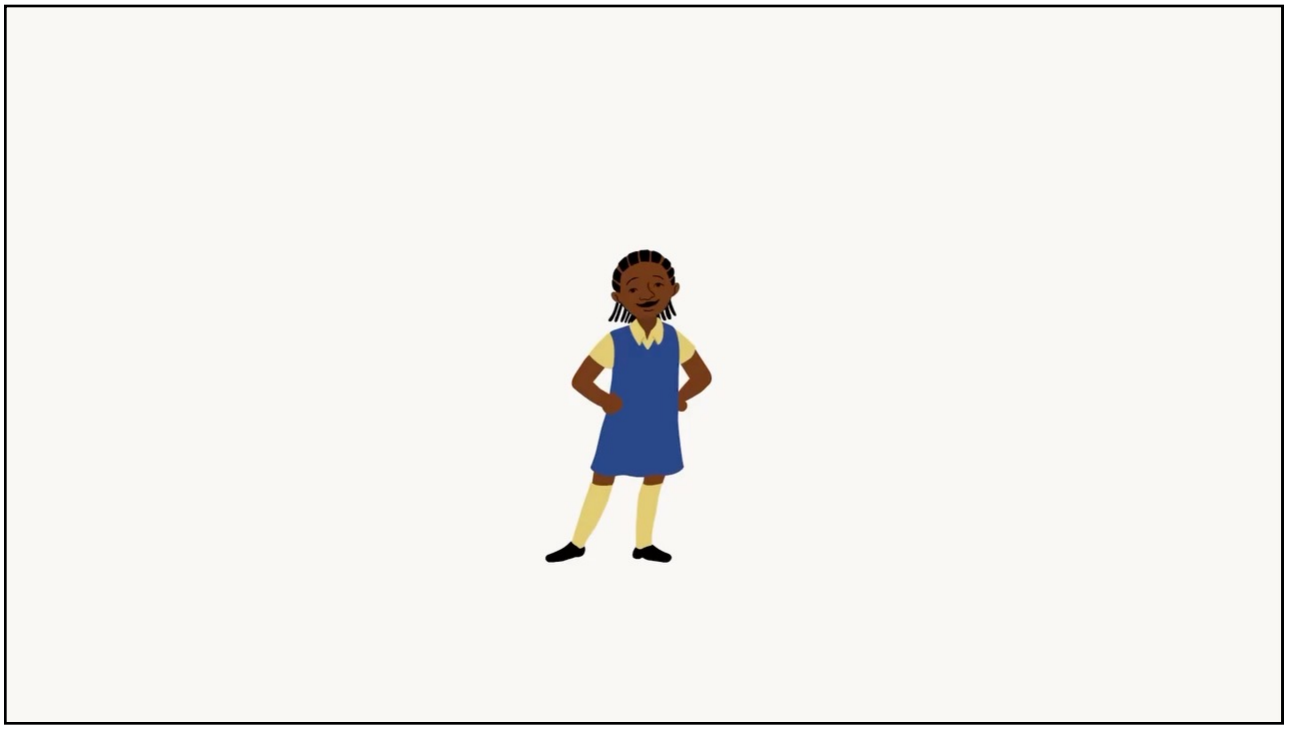


http://www.slido.com
event code **#P642**

2


THE SVERIGES RIKSBANK PRIZE
IN ECONOMIC SCIENCES IN MEMORY
OF ALFRED NOBEL 2023
Illustration: Niklas Elmehed
Claudia Goldin
"for having advanced our understanding
of women's labour market outcomes"
THE ROYAL SWEDISH ACADEMY OF SCIENCES
$
TIME
©Johan Jarnestad/The Royal Swedish Academy of Sciences
MARRIED WOMEN
IN WORK
%
60
40
20
0
AGRICULTURE
INDUSTRY
SERVICE
SOCIETY
CONTRACEPTIVE PILL
CHANGING EXPECTATIONS
EXPANDING EDUCATION
1790
1820
1850
1880
1910
1940
1970
YEAR
©Johan Jarnestad/The Royal Swedish Academy of Sciences

3



4

Administrative stuff

- Labs → vizitcards
- Upcoming assignment
  - Communicative visualization

5

VizItCards

Licia He and Eytan Adar, University of Michigan, 2020

6

What to look forward to…

- We're going to put you into small groups
  - Based on lab times
- You'll be responsible for finding 80 minutes
  - You can use the lab time
  - Needs to be synchronous
  - You'll be co-designing a solution
- At the end, we'll ask you to compare your solution to the "professional one"
  - Create up to 5 minute video (as a group)

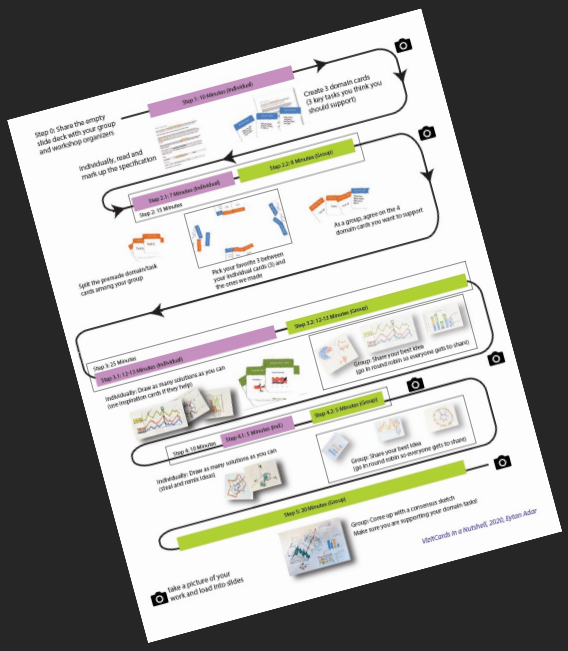
7

VizItCards Design Workshop

- The entire design process in 80-90 minutes
- Multi-phase
  - Declare goals
  - Produce many options
  - Bring them all together
- Clock driven
  - Time limits on every step
- Document as you go (google slides)

8

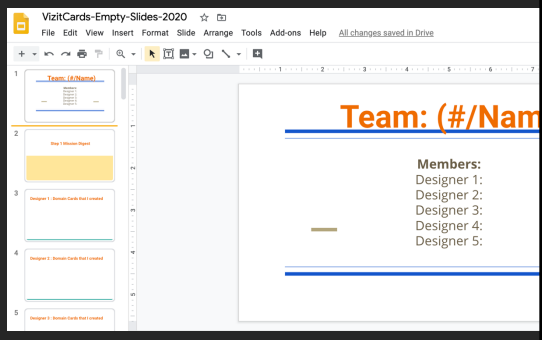
Don’t worry about following everything. We have a cheat sheet for you.



9

Step 0

- Introduce yourselves to your group!
- Make a copy of the Google slide deck and share it with
  - Everyone in your group
  - The team running the workshop

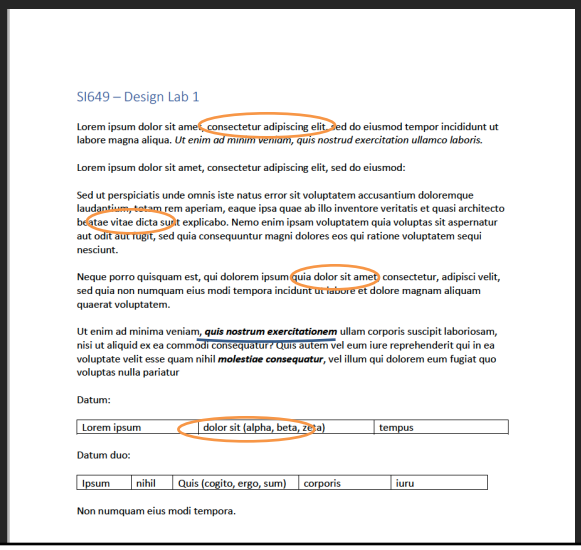


10

Step 1 – Digesting (10 minutes)

Individual

Read and markup the specification



SI649 – Design Lab 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore magna aliqua. *Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris.*

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod:

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt.

Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem.

Ut enim ad minima veniam, ***quis nostrum exercitationem*** ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil ***molestiae consequatur***, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur

Datum:

Lorem ipsum	dolor sit (alpha, beta, zeta)	tempus

Datum duo:

Ipsum	nihil	Quis (cogito, ergo, sum)	corporis	iuru

Non numquam eius modi tempora.

11

Step 1 – Digesting (10 minutes)

Individual

Create 3 domain cards



12



13

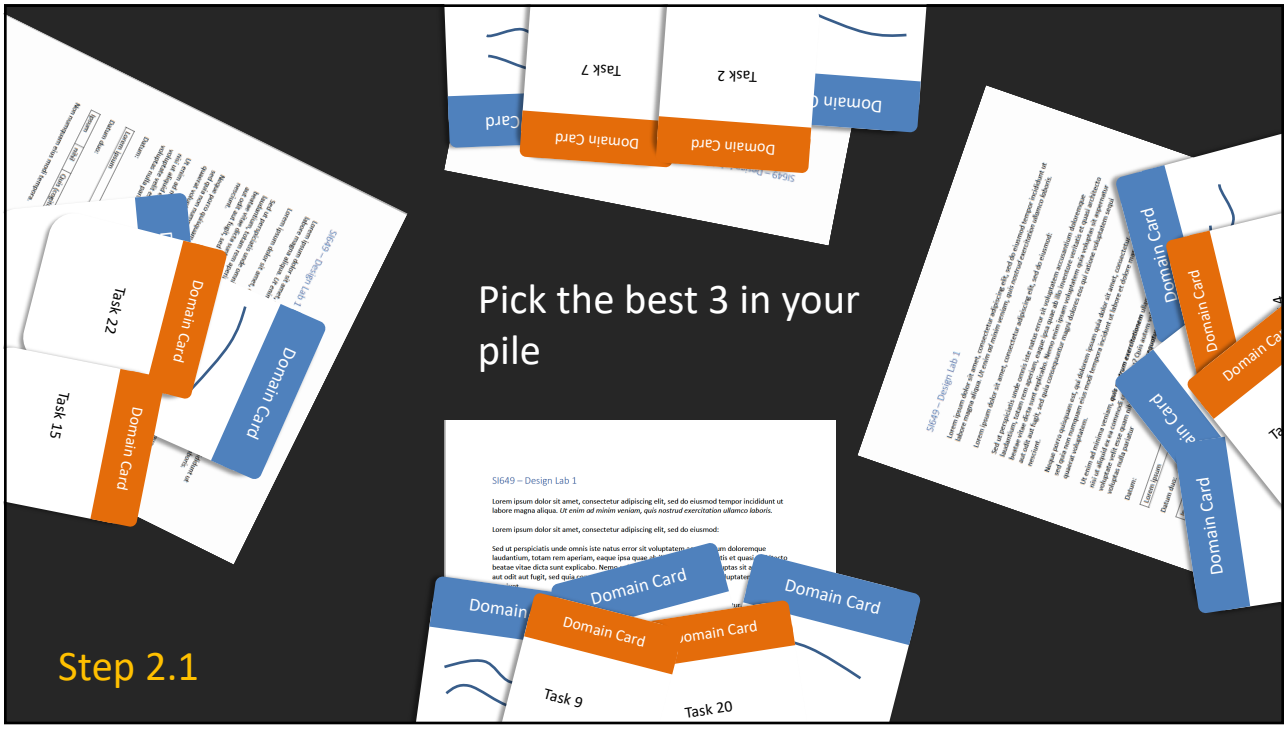
Step 2.1– Pick your favorites (7 minutes)

Individual

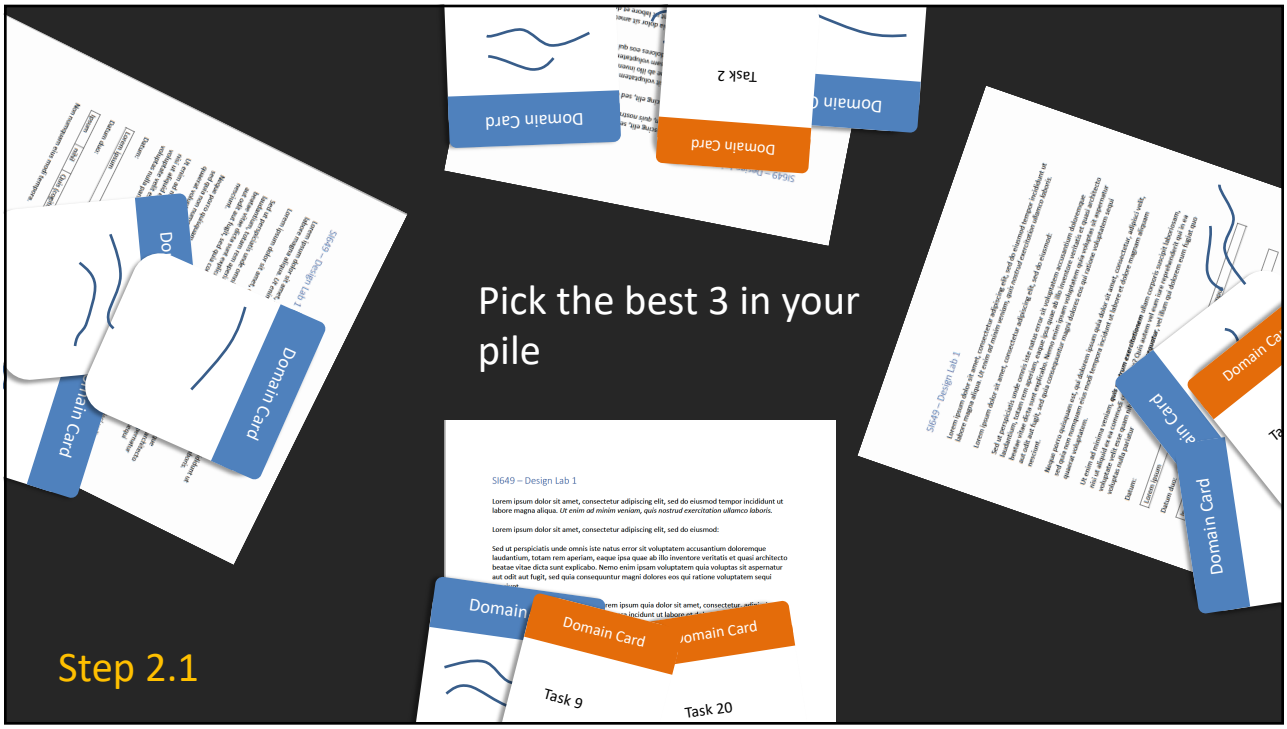
Split the deck we gave you equally



14


Pick the best 3 in your pile
Step 2.1
Task 22
Task 15
Task 7
Task 2
Task 9
Task 20
Domain Card
SI649 – Design Lab 1

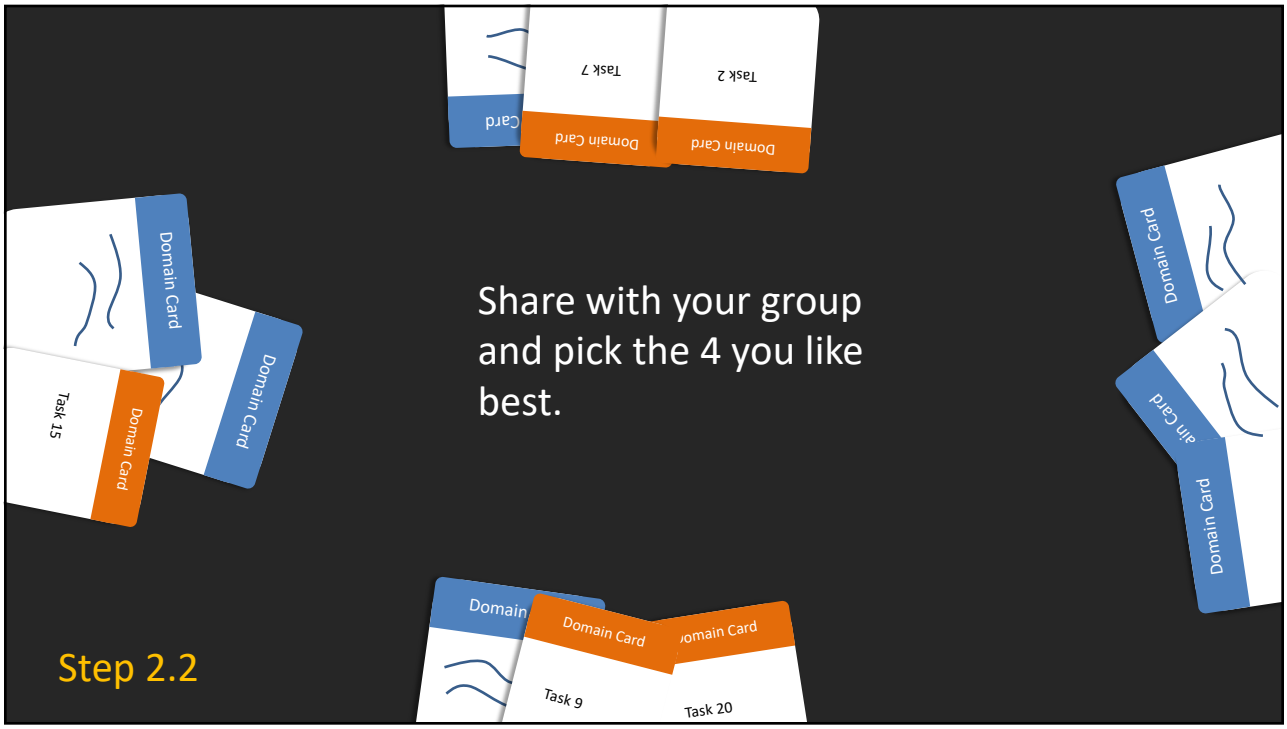
15


Pick the best 3 in your pile
Step 2.1
Task 2
Task 9
Task 20
Domain Card
SI649 – Design Lab 1

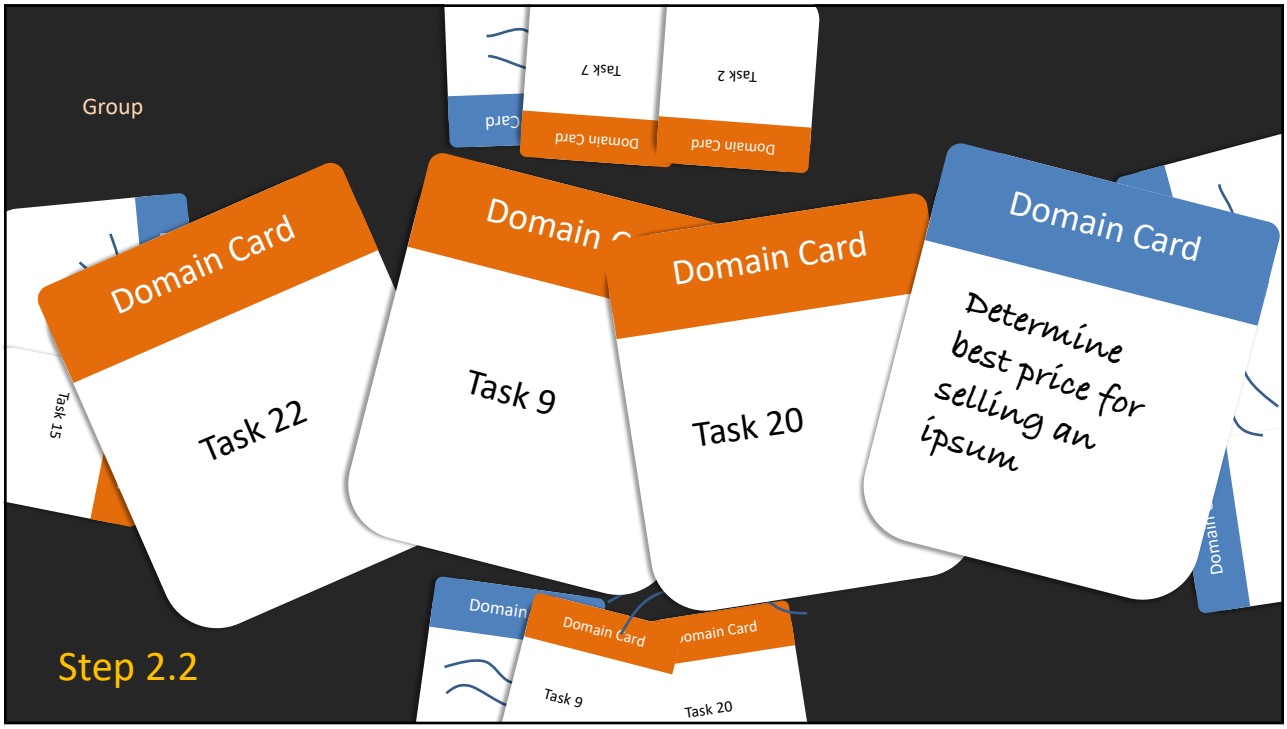
16


Step 2.2– Find the best 4 (8 minutes)
Group
Domain Card
Task 22
Domain Card
Task 9
Domain Card
Task 20
Domain Card
Determine best price for selling an ipsum

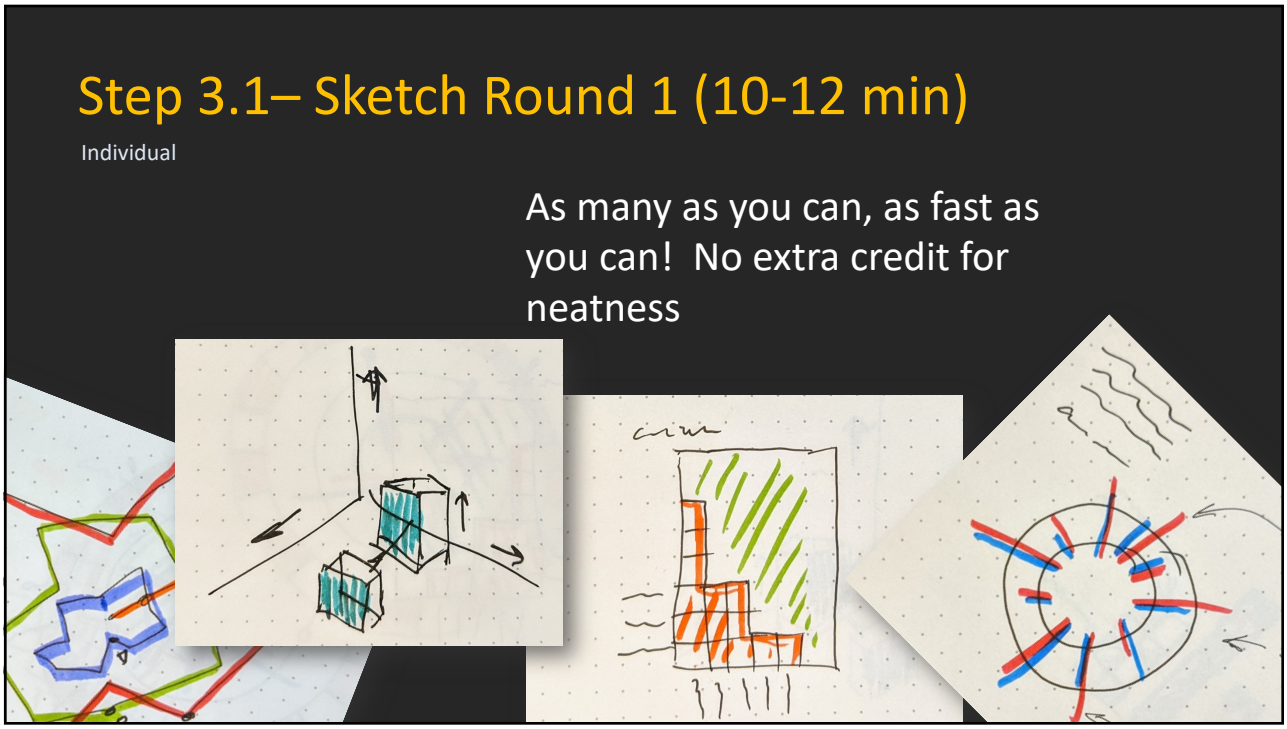
17


Share with your group and pick the 4 you like best.
Step 2.2
Domain Card
Task 15
Domain Card
Domain Card
Domain Card
Task 9
Domain Card
Task 20

18


Group
Domain Card
Task 22
Task 9
Domain Card
Task 20
Domain Card
Determine best price for selling an ipsum
Step 2.2

19


Step 3.1– Sketch Round 1 (10-12 min)
Individual
As many as you can, as fast as you can! No extra credit for neatness

20

Step 3.1– Sketch Round 1 (10-12 min)

Individual

We'll give you "inspiration" cards for help



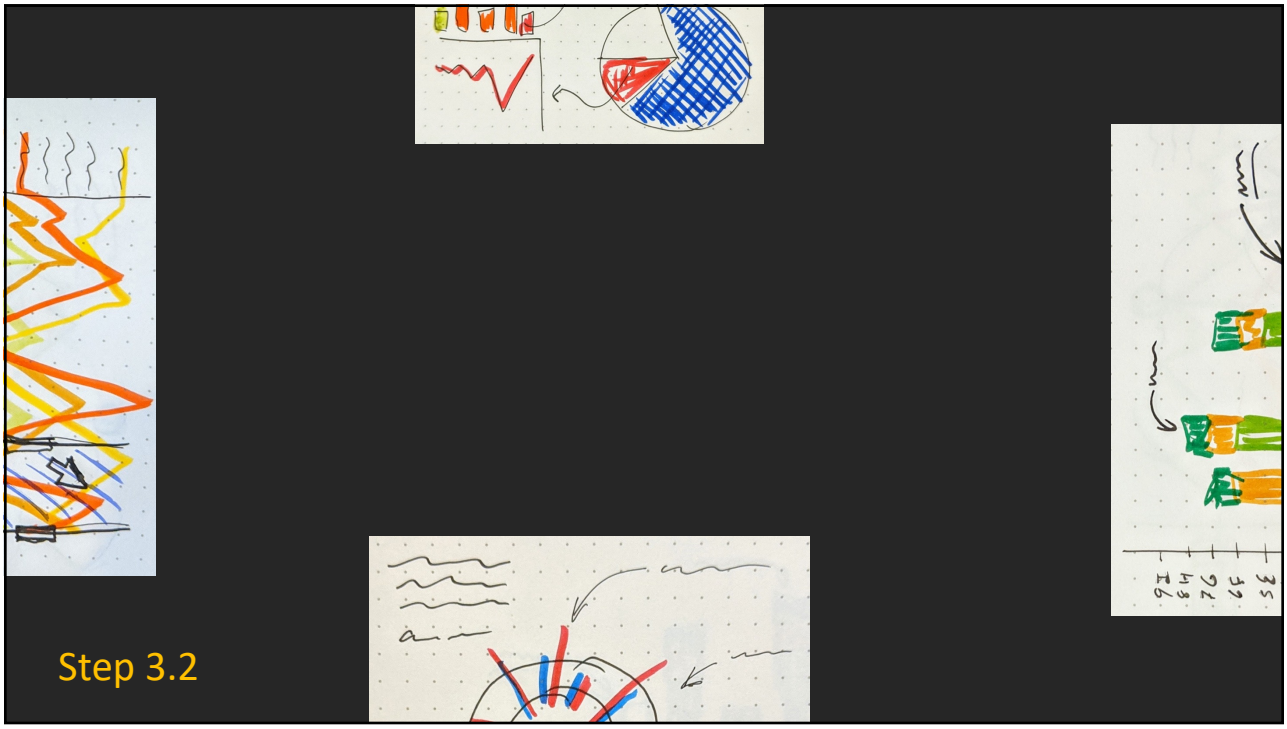
21

Step 3.2– Sketch Round 1 (10 min)

Group

Share your best idea(s)

22

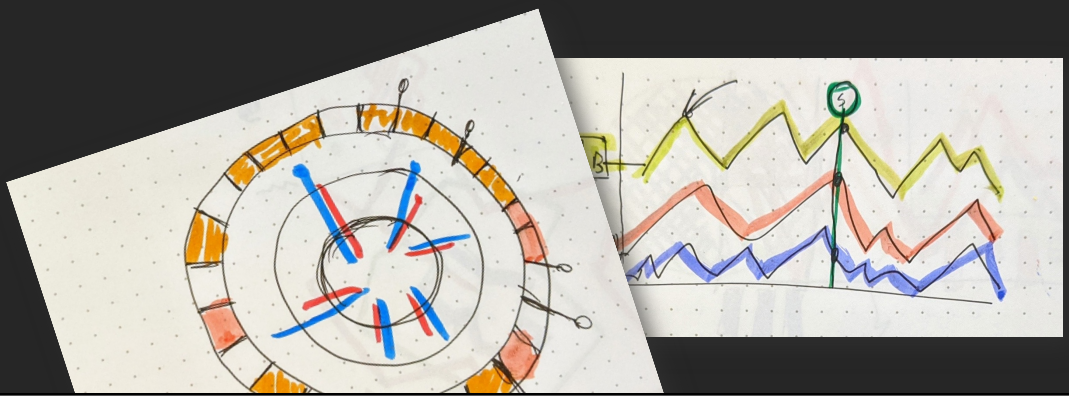


23

Step 4.1– Sketch Round 2 (5 min)

Individual

Produce a new set (steal and recombine from your teammates). Fast and messy is good!



24

Step 4.2– Sketch Round 1 (5 min)

Group

Share your best idea(s) again

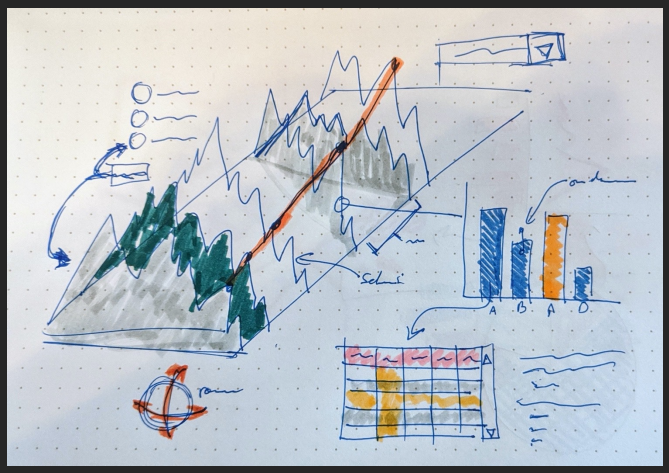
25



26

Step 5 – Final Sketch (20 minutes)

Group



27

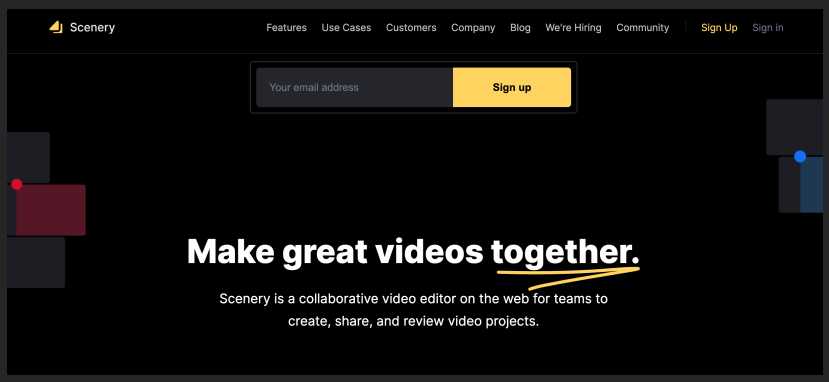
A few things

- Make sure you are meeting your domain card goals in every step
  - You picked them!
- Document as you go
  - take pictures of intermediate products and upload to the shared Google slides
- Take turns and be polite
- Be aware of time
  - we don't want you go over, but we do want you to finish

28

Video Report

- Optional, but really useful tool for making your presentations (http://scenery.video):



29

Upcoming project

- Food!

- Phase 1 – you'll create a static infographic
  - You pick the objectives (consistent with article)
  - It should be functional but also look good
- Phase 1 – you'll revise your infographic (if you want) and/or create an interactive version (for extra credit)

30

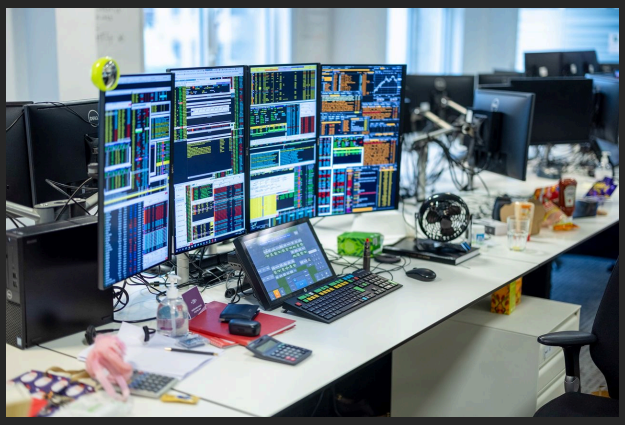
Back to our program…

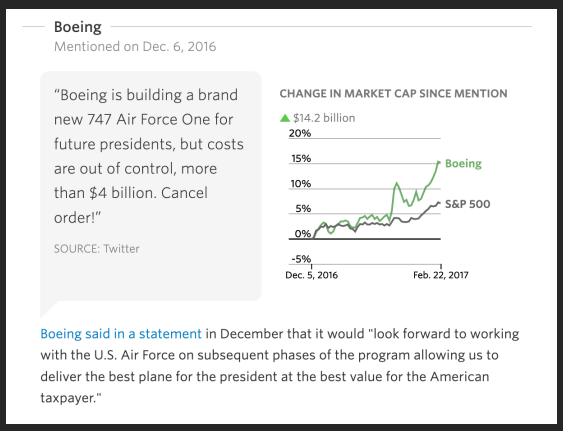
31

High level functions

- Record
- Analyze
- Communicate

32





33

Rule of thumb

- If you have a hypothesis you are probably generating analytical visualizations
  - Your objective is finding an answer/building patterns/making decision/etc.
- If you are trying to share your answer/patterns/decision with someone else you are generating a communicative visualization
  - Your objective is to make sure the other person learned whatever you had to teach them

34

You are…	Your client is…

35

	You are…	Your client is…
Someone working for a school, a game company, a trading firm, etc.	An analyst making visualizations	Yourself (but often guided by needs of an organization)
Tableau, but also someone making a dashboard for the network analysts at TMobile	Someone making visualization software for analysts	A lot of *different* analysts, making visualizations to answer *many* questions
Adobe, LineUp, etc.	Someone making visualization software for communicators/ designers	A lot of designers, with many *different* "expressive" needs
Graphic designer at the NYTimes, but also an analyst who needs to present their findings to their company	A communicator/ designer	Your viewers (and probably who is paying your bills)

36

Just for fun… some questions about structure

http://www.slido.com
event code **#P642**

37

Emphasis and control



https://www.youtube.com/watch?v=v4seDVfgwOg

38





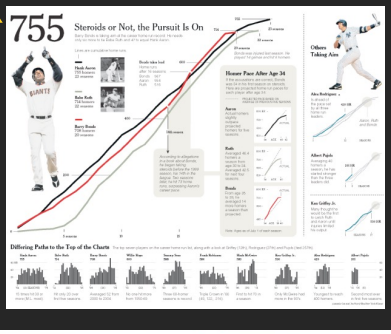
Montage / editing
- Prescribed order
- Where you look explicitly directed
- Little reader/viewer agency

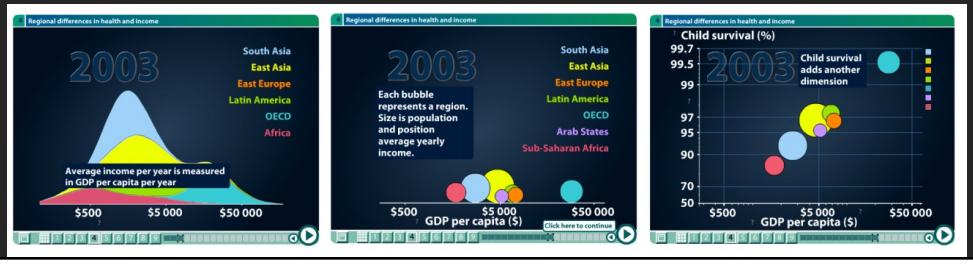
Ensemble staging / direction
- Less prescribed order
- Salience, structure nudges you to look
- More reader/viewer agency

39

Visual narrative tactics

- Highlighting
- Transition guidance

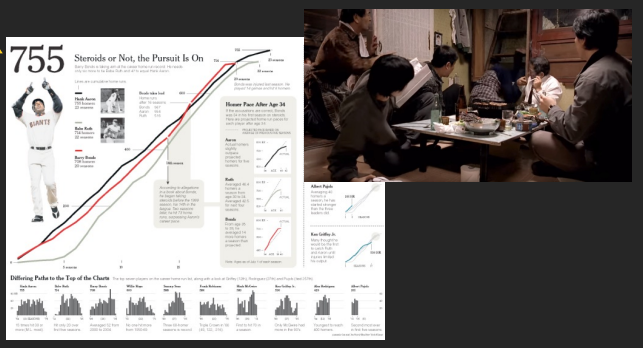


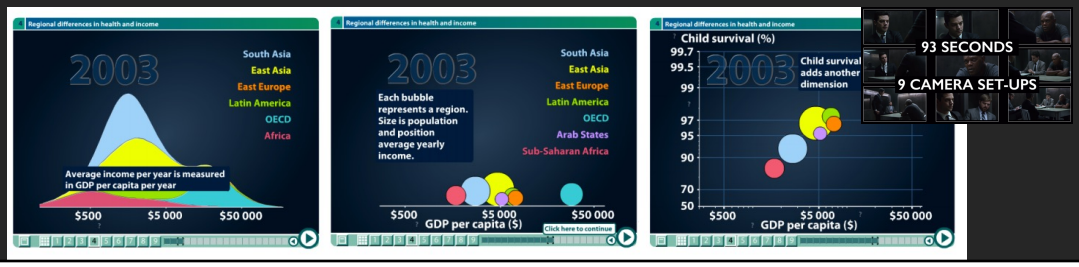


40

Visual narrative tactics

- Highlighting
- Transition guidance





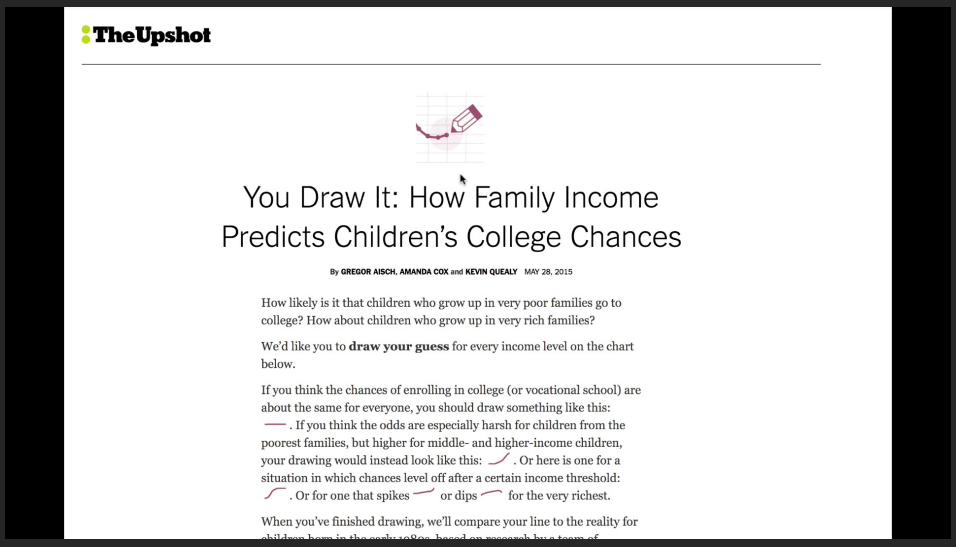
41

Learning Objectives Framework

- Paper from 2021 shows the relative benefits of "specifying" visualizations through LOs, insights, and assessments
- New paper on affective learning objectives
- Other ongoing work:
  - Tools to help build assessments
- But…
  - We've been having students use this idea for a while (you will too!)

43

Draw Your Expectations…



The Upshot

You Draw It: How Family Income Predicts Children's College Chances

By GREGOR AISCH, AMANDA COX and KEVIN QUEALY MAY 28, 2015

How likely is it that children who grow up in very poor families go to college? How about children who grow up in very rich families?

We'd like you to **draw your guess** for every income level on the chart below.

If you think the chances of enrolling in college (or vocational school) are about the same for everyone, you should draw something like this: —. If you think the odds are especially harsh for children from the poorest families, but higher for middle- and higher-income children, your drawing would instead look like this: ⌒. Or here is one for a situation in which chances level off after a certain income threshold: ⌒. Or for one that spikes ⌒ or dips ⌒ for the very richest.

When you've finished drawing, we'll compare your line to the reality for

44

Group question: What might be the advantages of "draw your own?"

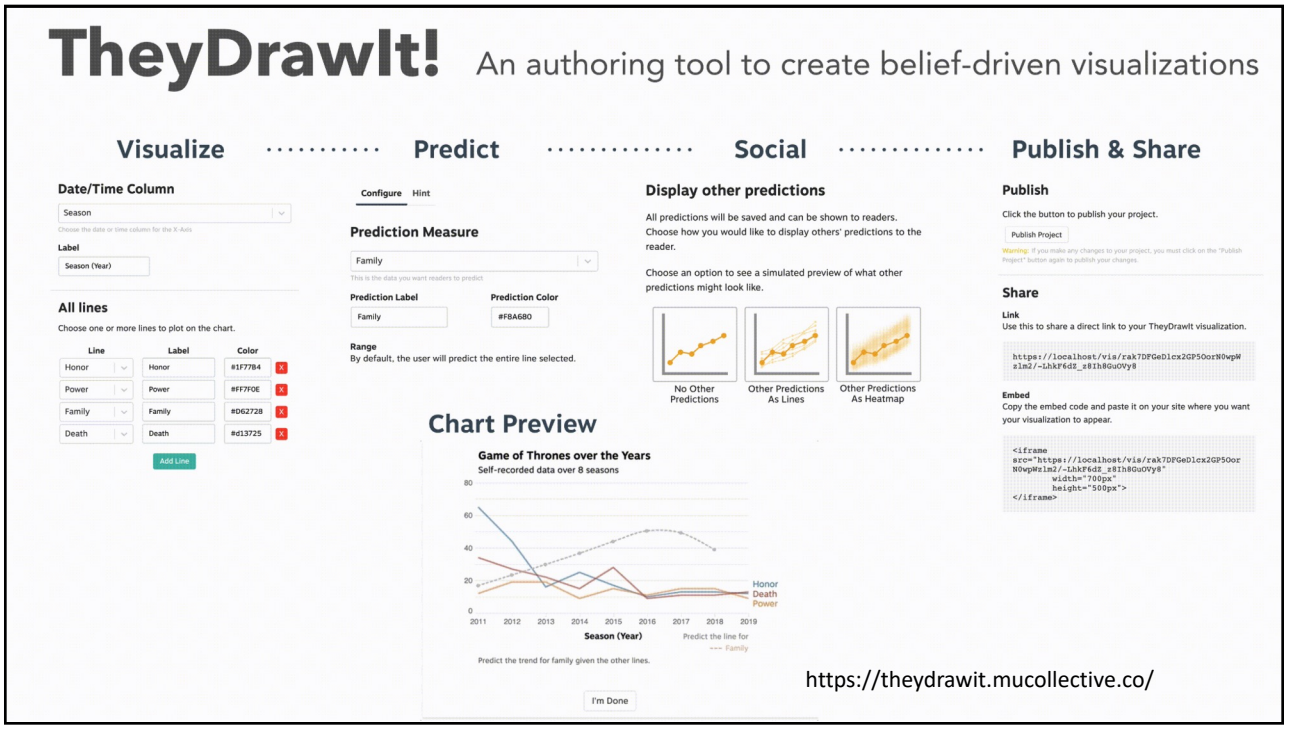
http://www.slido.com
event code **#P642**

45

Draw Your Expectations

- Recall
  - 20-25% improvement in accuracy when you draw your own
  - No impact if you only use text
  - Similar improvement if you see other people's drawn expectations

46



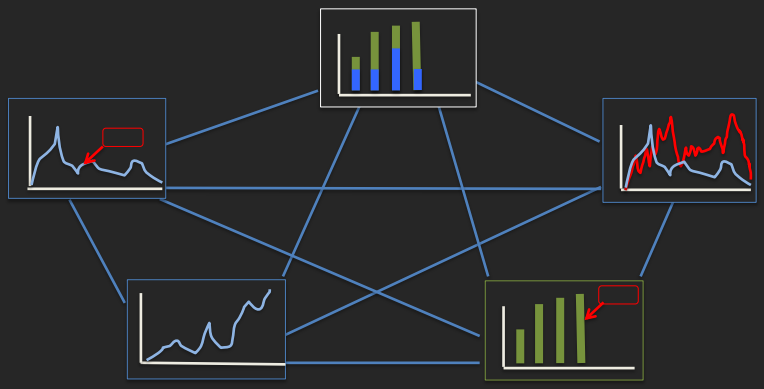
https://theydrawit.mucollective.co/

47

Graph-based approach

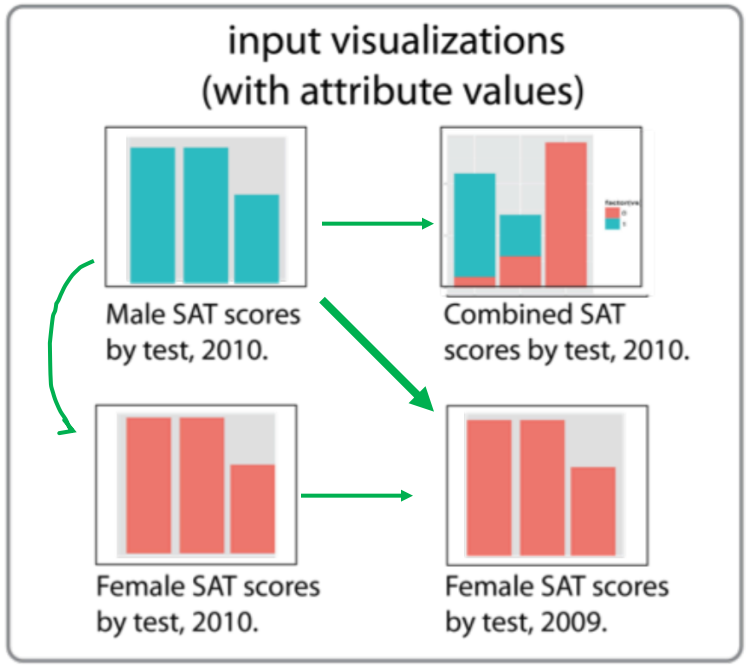
Find paths that:

- Have a low cost,



48

A Deeper Understanding of Sequence in Narrative Visualization Jessica Hullman, Steven Drucker, Nathalie Henry Riche, Bongshin Lee, Danyel Fisher, and Eytan Adar



49

Representation	Relevant Transition Types	Relevant Interactions
Dependent variable	Comparative – Measure walk	Sort, Derive, Navigate (Distortion), Coordinate (small multiples)
Independent variable	Comparative – Dimension walk	Filter (independent variable, such as with query widget), Navigate (scroll, pan), Coordinate (small multiples)
Time	Temporal	Filter (direct selection, slider), Coordinate (small multiples)
Hierarchical relation	Granularity	Filter (direct selection, query widget, slider), Navigate (overview & detail, zoom, semantic zoom), Derive (aggregate)

50

Draco: Representing, Applying & Learning Visualization Design Guidelines

UW Interactive Data Lab Follow
Oct 22, 2018 · 9 min read

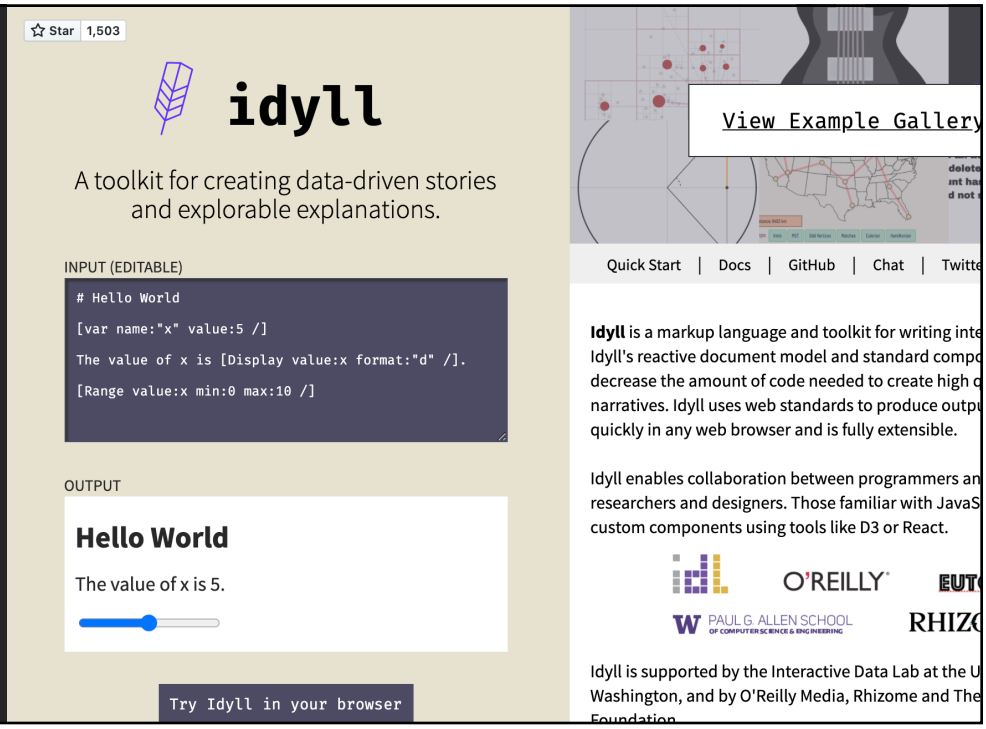
From academic courses to online articles, discussions of visualization often abound with design guidelines: *Don't use pie charts! Don't use rainbow color maps! Ensure axes include zero!* When applied in a proper context, such guidelines help prevent misleading visualizations. However, such guidelines are not always well-known, and are themselves subject to debate among practitioners, in part because the "proper context" may not be obvious.

As a result, design guidelines can be challenging to apply, relying on people's judgment and expertise. What guidelines does my design adhere to? Are there exceptions to the "rules"? For example, we are often told a

https://uwdata.github.io/draco/

51

https://idyll-lang.org/



52

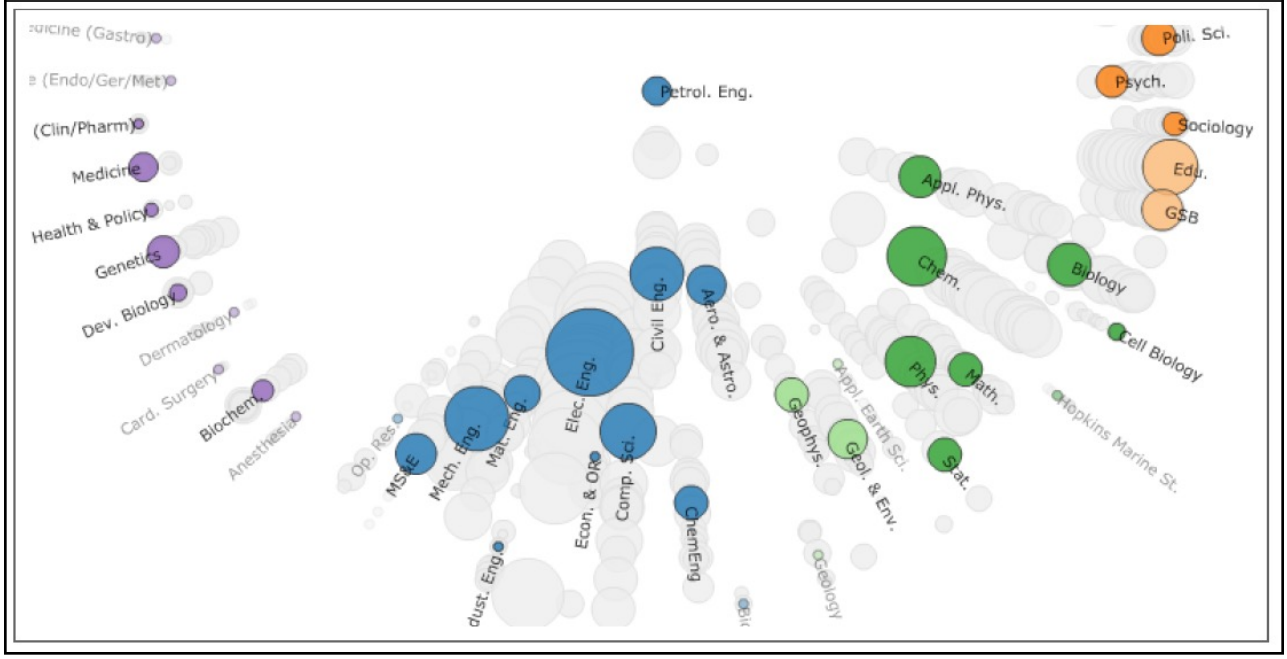
Anti-objectives (preventing the viewer from misinterpreting the data)

53



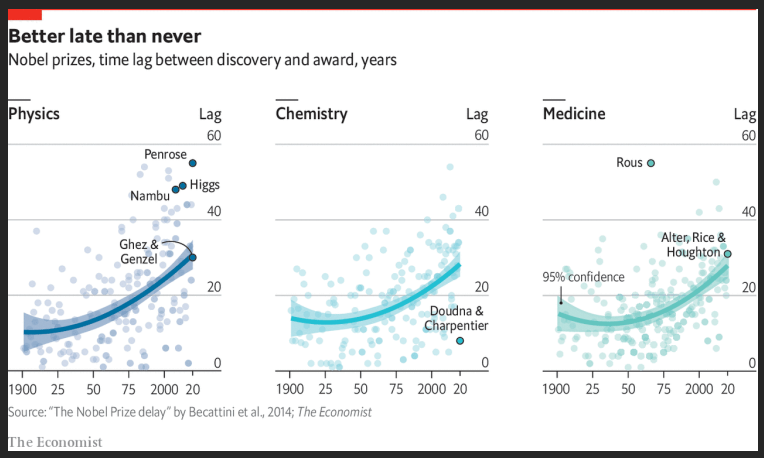
http://www.slido.com
event code **#P642**

54



55

Something to keep in mind: most communicative visualization isn't really controversial



56

Group question: What is our responsibility as vis designers to our viewers? Should we take a paternalistic role?

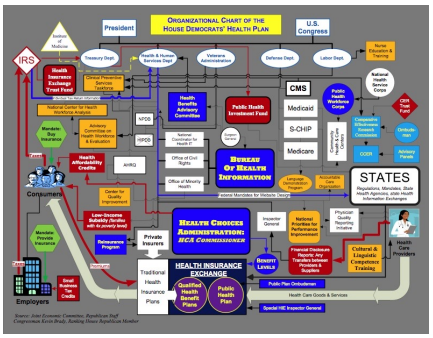
http://www.slido.com
event code **#P642**

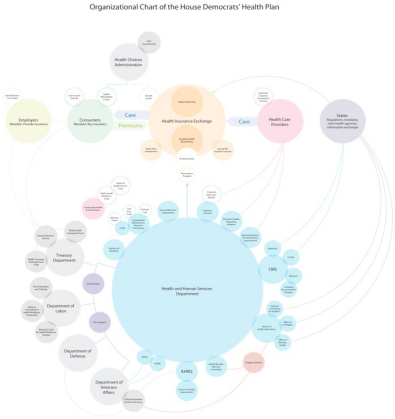
57

The politics of visualization…

58

Visual metaphor (mapping rhetoric)

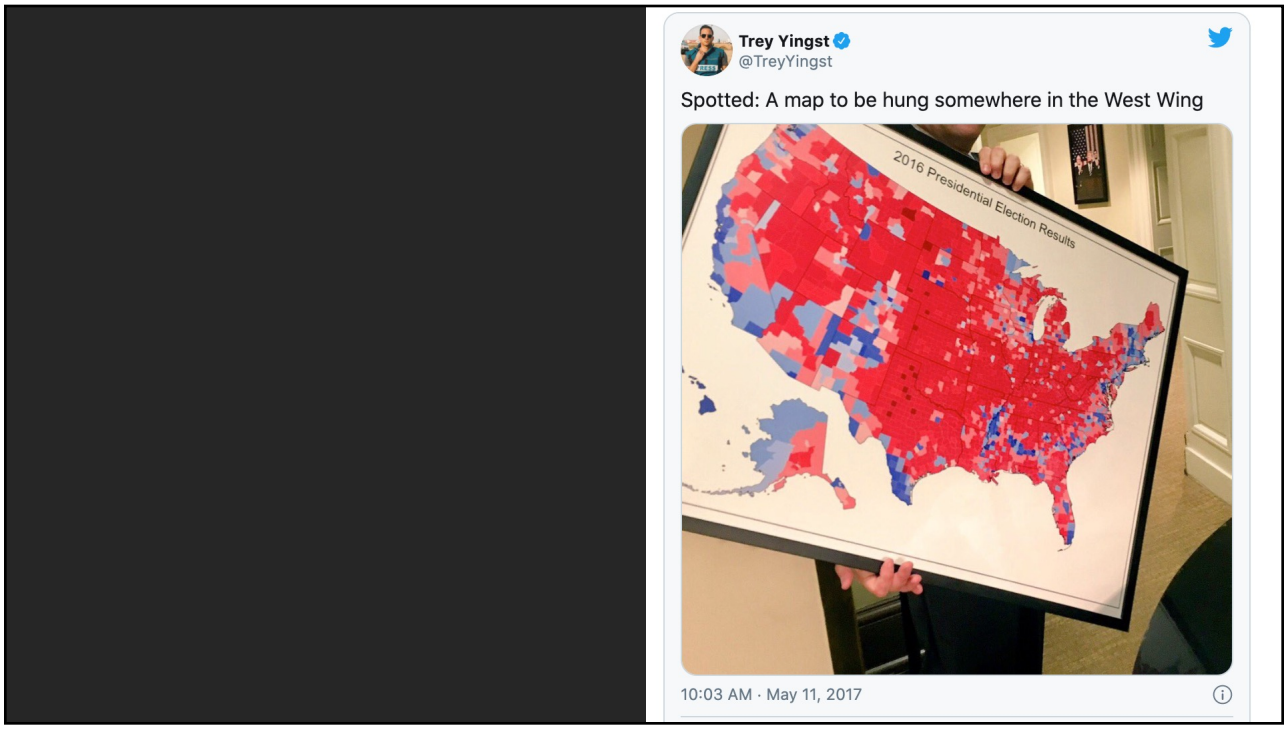




59


Lara Trump
@LaraLeaTrump
Try to impeach this.
3:36 PM · Sep 28, 2019 from Briarcliff Manor, NY · Twitter for iPhone

60


Trey Yingst
@TreyYingst
Spotted: A map to be hung somewhere in the West Wing
2016 Presidential Election Results
10:03 AM · May 11, 2017

61



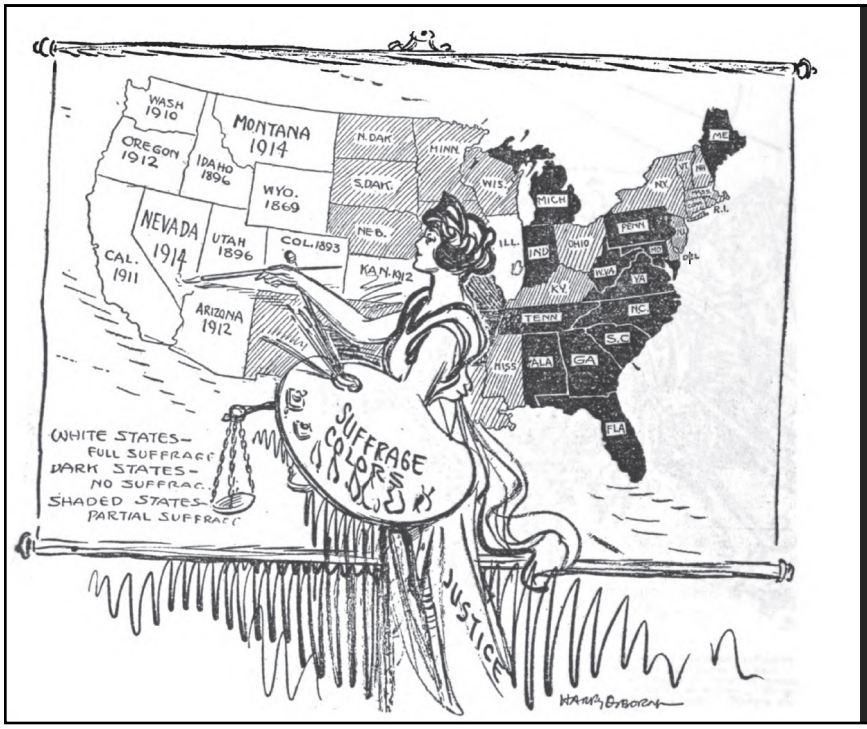
https://observablehq.com/@karimdouieb/try-to-impeach-this-challenge-accepted

62

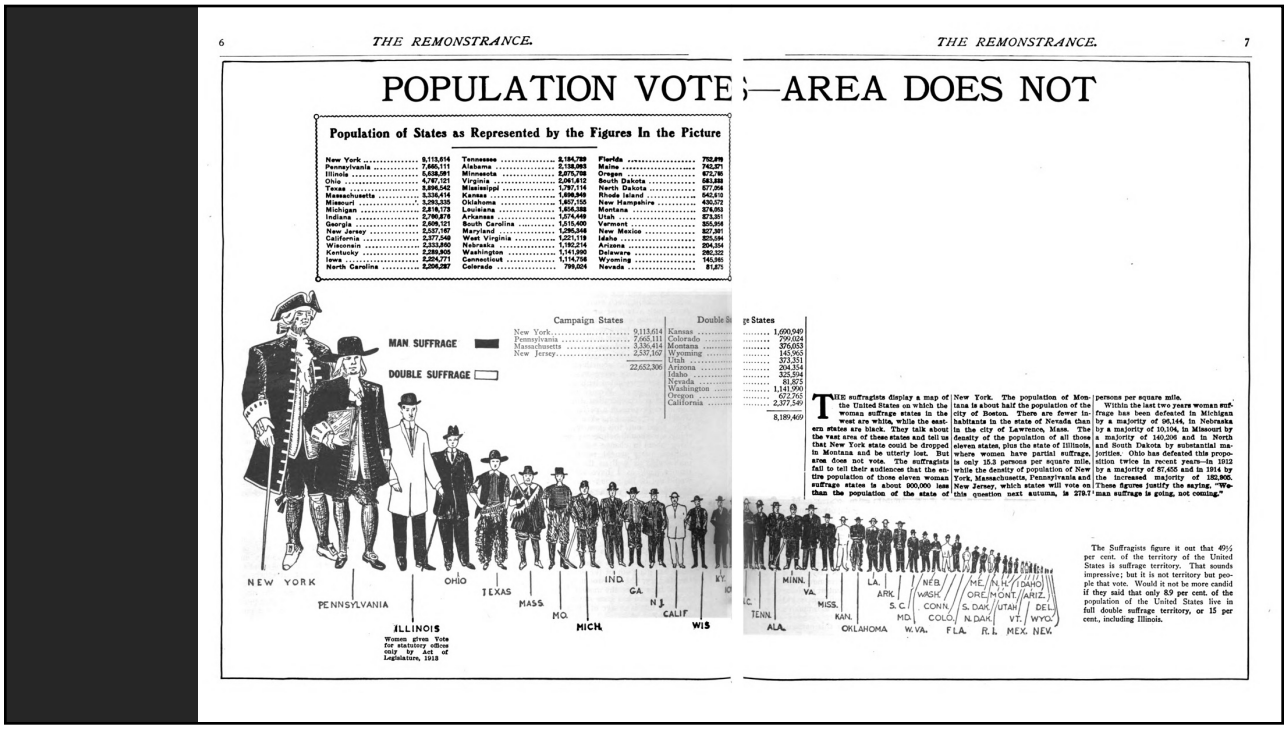
DO YOU KNOW that in all the white territory in Canada and the United States on the map on next page women vote on equal terms; in the grey territory women have some suffrage privileges and in the black none? Note that the black territory is Central America, the frozen North, and the very conservative East and South. **Women vote in nearly half the territory of the United States.**



63



64



POPULATION VOTES—AREA DOES NOT

Population of States as Represented by the Figures In the Picture

New York	9,113,614	Tennessee	2,184,789	Florida	752,619
Pennsylvania	7,665,111	Alabama	2,138,093	Maine	742,371
Illinois	5,638,591	Minnesota	2,075,708	Oregon	672,765
Ohio	4,767,121	Virginia	2,061,612	South Dakota	583,888
Texas	3,896,542	Mississippi	1,797,114	North Dakota	577,056
Massachusetts	3,336,416	Kansas	1,690,949	Rhode Island	542,610
Missouri	3,293,335	Oklahoma	1,657,155	New Hampshire	430,572
Michigan	2,810,173	Louisiana	1,656,388	Montana	376,053
Indiana	2,700,876	Arkansas	1,574,449	Utah	373,351
Georgia	2,609,121	South Carolina	1,515,400	Vermont	355,956
New Jersey	2,537,167	Maryland	1,295,346	New Mexico	327,301
California	2,377,549	West Virginia	1,221,119	Idaho	325,594
Wisconsin	2,333,860	Nebraska	1,192,214	Arizona	204,354
Kentucky	2,289,905	Washington	1,141,990	Delaware	202,322
Iowa	2,224,771	Connecticut	1,114,756	Wyoming	145,965
North Carolina	2,206,287	Colorado	799,024	Nevada	81,875

Campaign States		Double Suffrage States	
New York	9,113,614	Kansas	1,690,949
Pennsylvania	7,665,111	Colorado	799,024
Massachusetts	3,336,414	Montana	376,053
New Jersey	2,537,167	Wyoming	145,965
	22,652,306	Utah	373,351
		Arizona	204,354
		Idaho	325,594
		Nevada	81,875
		Washington	1,141,990
		Oregon	672,765
		California	2,377,549
			8,189,469

THE suffragists display a map of the United States on which the woman suffrage states in the west are white, while the eastern states are black. They talk about the vast area of these states and tell us that New York state could be dropped in Montana and be utterly lost. But area does not vote. The suffragists fail to tell their audiences that the entire population of those eleven woman suffrage states is about 900,000 less than the population of the state of New York. The population of Montana is about half the population of the city of Boston. There are fewer inhabitants in the state of Nevada than in the city of Lawrence, Mass. The density of the population of all those eleven states, plus the state of Illinois, where women have partial suffrage, is only 15.3 persons per square mile, while the density of population of New York, Massachusetts, Pennsylvania and New Jersey, which states will vote on this question next autumn, is 270.7 persons per square mile.

Within the last two years woman suffrage has been defeated in Michigan by a majority of 96,144, in Nebraska by a majority of 10,104, in Missouri by a majority of 140,206 and in North and South Dakota by substantial majorities. Ohio has defeated this proposition twice in recent years—in 1912 by a majority of 87,455 and in 1914 by the increased majority of 152,906. These figures justify the saying, "Woman suffrage is going, not coming."

The Suffragists figure it out that 49½ per cent. of the territory of the United States is suffrage territory. That sounds impressive; but it is not territory but people that vote. Would it not be more candid if they said that only 8.9 per cent. of the population of the United States live in full double suffrage territory, or 15 per cent., including Illinois.

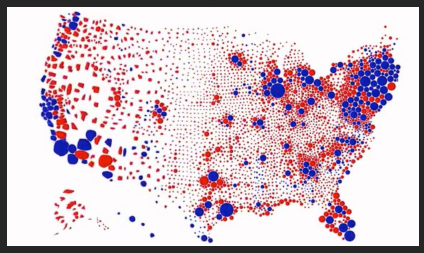
65

It's the area that counts

It's the population that counts

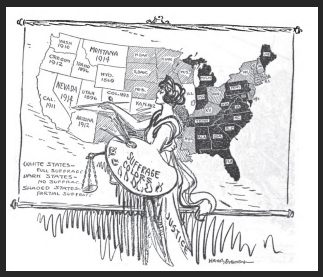
Conservative view is popular

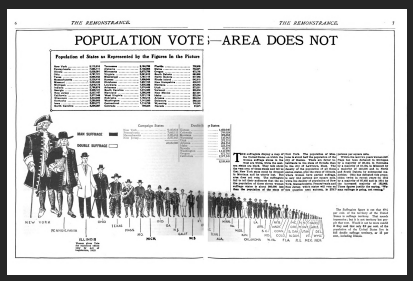




Liberal

Liberal





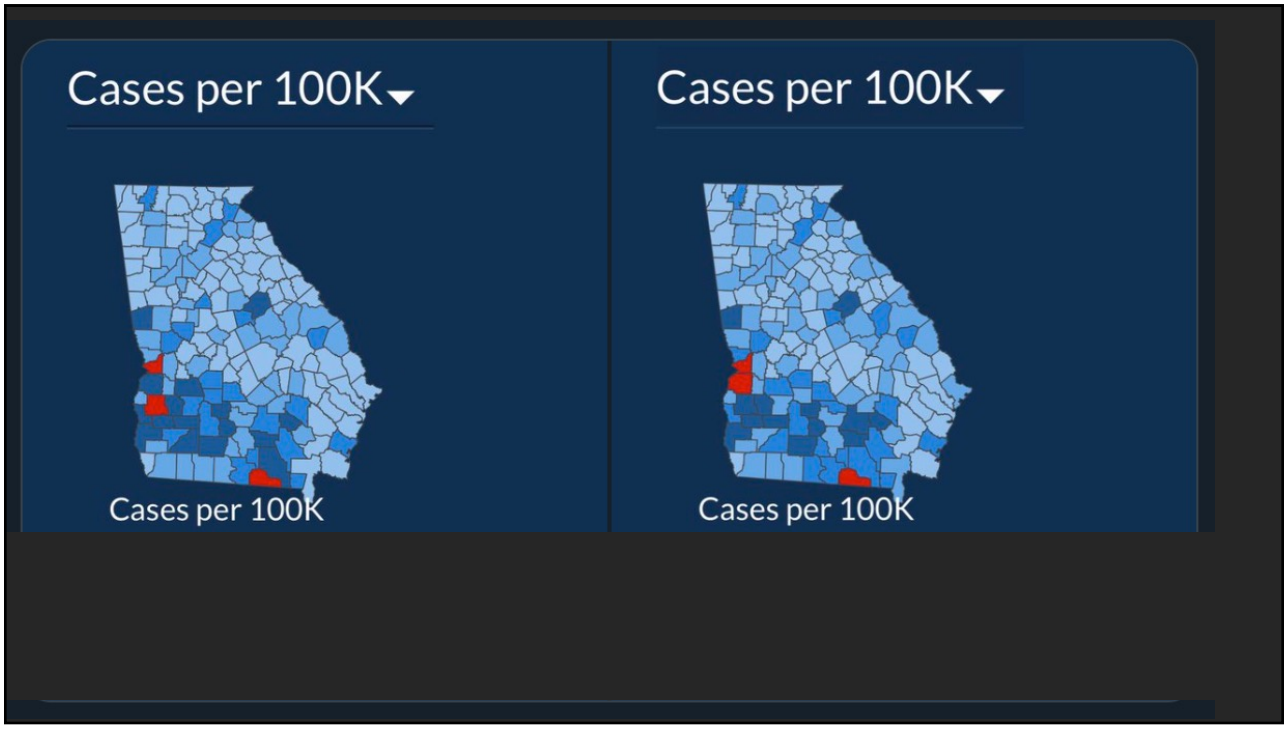
Conservative

66

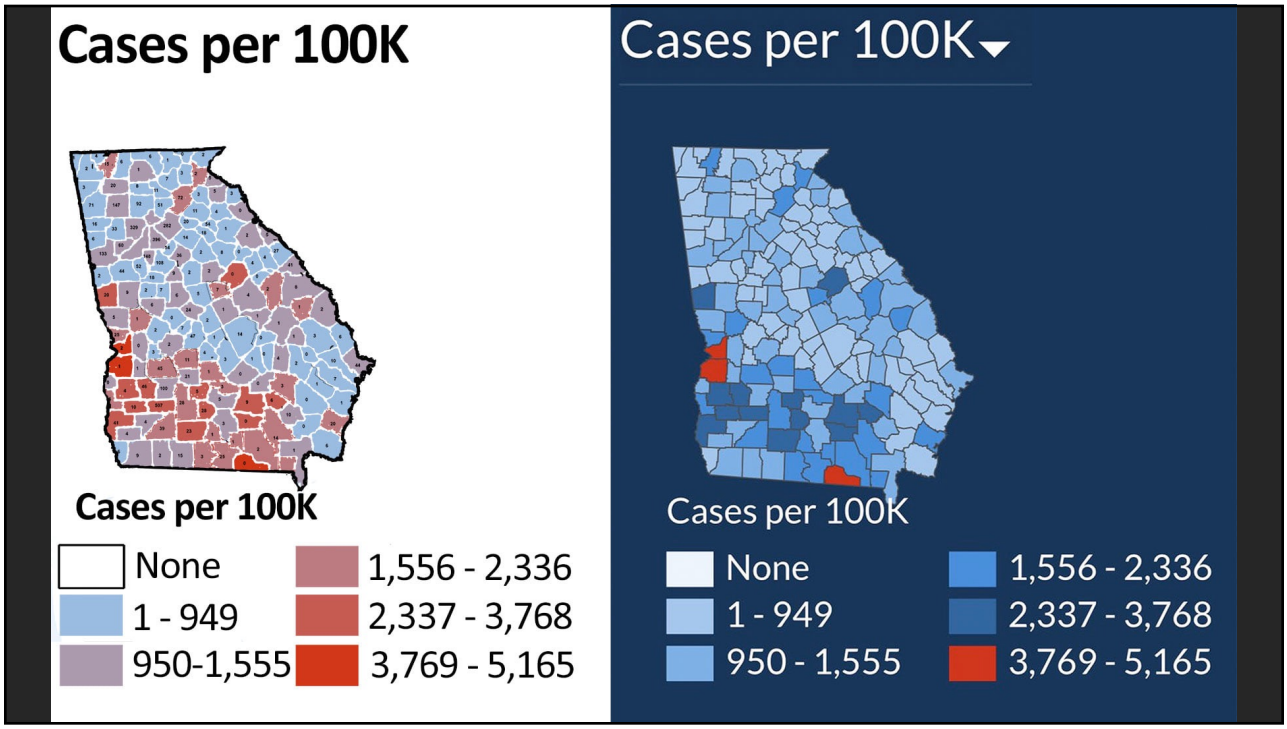
Group question: How do we think about visualization ethics in the context of these two examples? Can/should visualizations be non-political? Are they equivalent?

http://www.slido.com
event code **#P642**

67


Cases per 100K
Cases per 100K
Cases per 100K
Cases per 100K
Cases per 100K

68


Cases per 100K
Cases per 100K
None
1 - 949
950-1,555
1,556 - 2,336
2,337 - 3,768
3,769 - 5,165
Cases per 100K
Cases per 100K
None
1 - 949
950 - 1,555
1,556 - 2,336
2,337 - 3,768
3,769 - 5,165

69

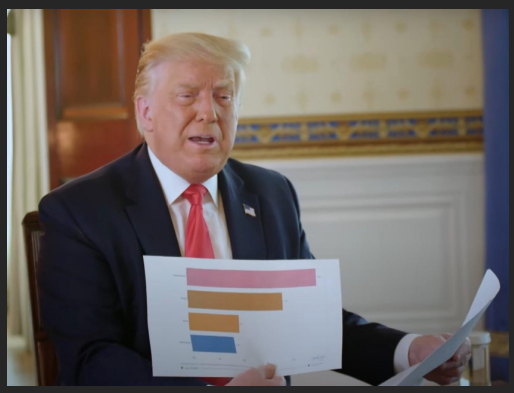
Was this…

- An act of deception?
- An honest mistake?
- Something else?

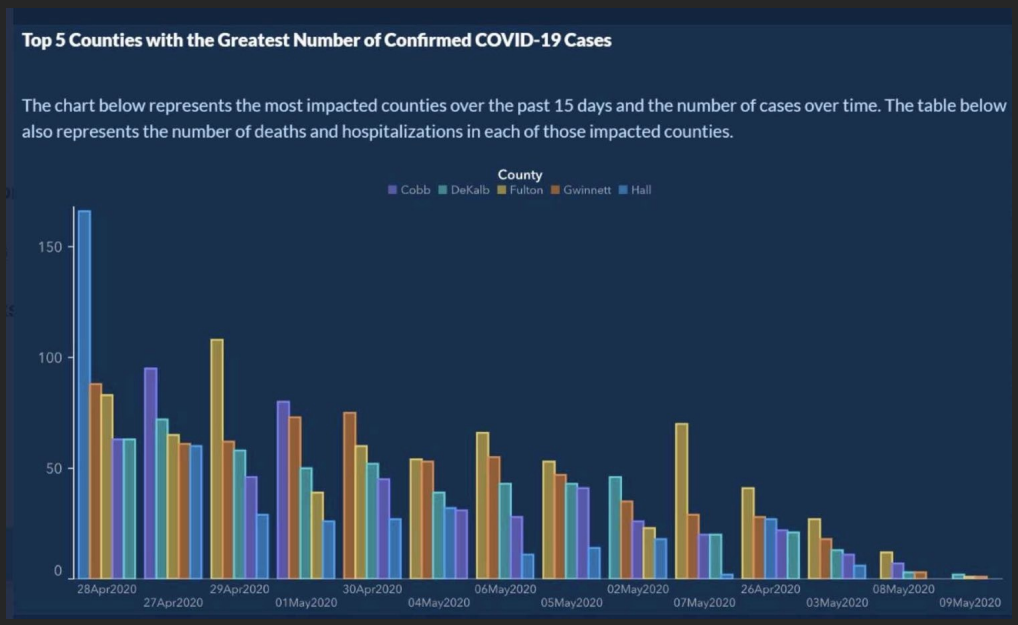
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70





71

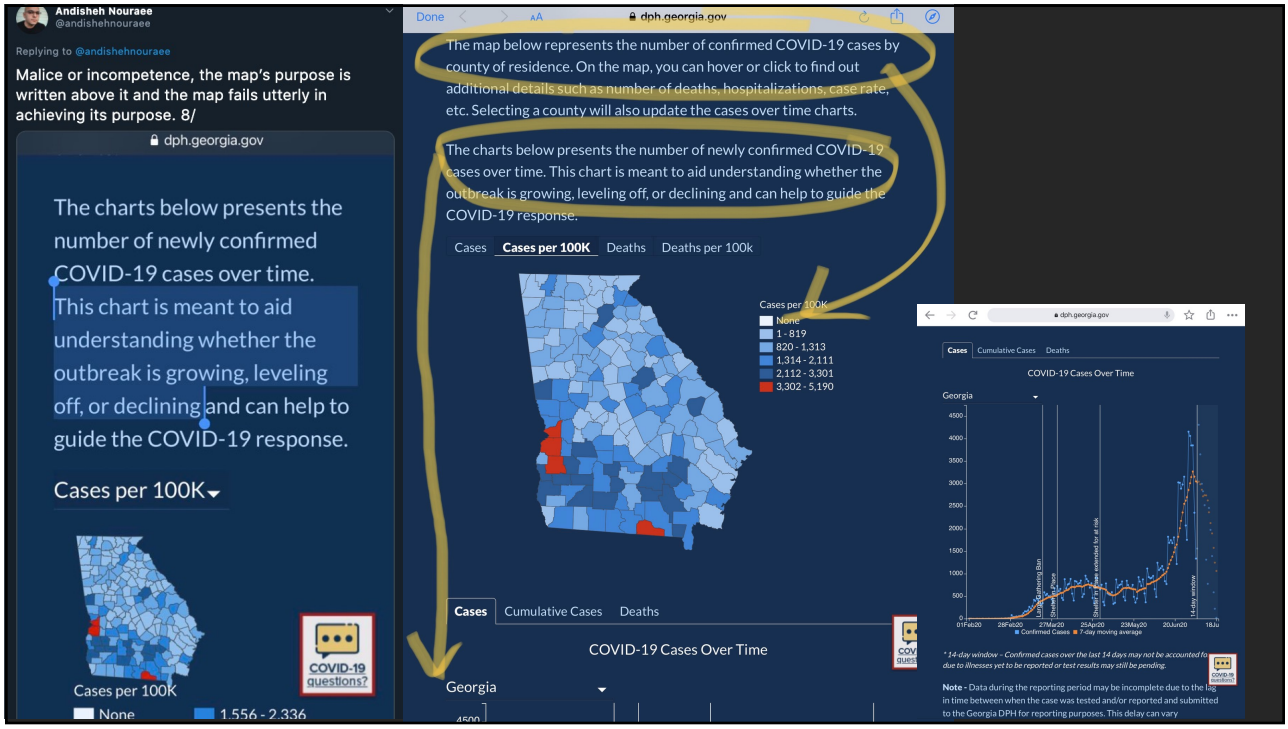


72

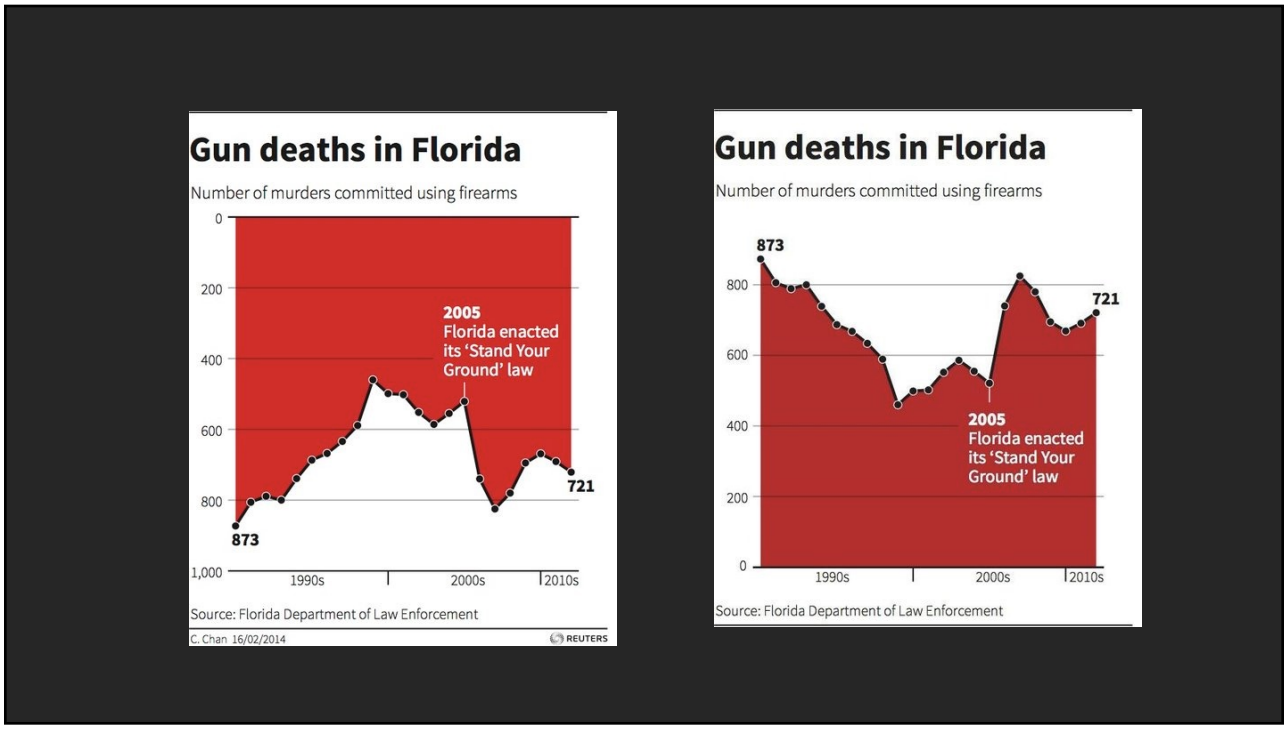
Was this…

- An act of deception?
- An honest mistake?
- An honest mistake that wasn't caught because it fit the story they wanted to tell?

73


Andisheh Nouraee
@andishehnouraee
Replying to @andishehnouraee
Malice or incompetence, the map's purpose is written above it and the map fails utterly in achieving its purpose. 8/
dph.georgia.gov
The charts below presents the number of newly confirmed COVID-19 cases over time. This chart is meant to aid understanding whether the outbreak is growing, leveling off, or declining and can help to guide the COVID-19 response.
Cases per 100K
Cases per 100K
None
1,556 - 2,336
COVID-19 questions?
Done
dph.georgia.gov
The map below represents the number of confirmed COVID-19 cases by county of residence. On the map, you can hover or click to find out additional details such as number of deaths, hospitalizations, case rate, etc. Selecting a county will also update the cases over time charts.
The charts below presents the number of newly confirmed COVID-19 cases over time. This chart is meant to aid understanding whether the outbreak is growing, leveling off, or declining and can help to guide the COVID-19 response.
Cases
Cases per 100K
Deaths
Deaths per 100k
Cases per 100K
None
1 - 819
820 - 1,313
1,314 - 2,111
2,112 - 3,301
3,302 - 5,190
Cases
Cumulative Cases
Deaths
COVID-19 Cases Over Time
Georgia
Cases
Cumulative Cases
Deaths
COVID-19 Cases Over Time
Georgia
Confirmed Cases
7-day moving average
Note - Data during the reporting period may be incomplete due to the lag in time between when the case was tested and/or reported and submitted to the Georgia DPH for reporting purposes. This delay can vary

74


Gun deaths in Florida
Number of murders committed using firearms
0
200
400
600
800
1,000
2005
Florida enacted its 'Stand Your Ground' law
873
721
1990s
2000s
2010s
Source: Florida Department of Law Enforcement
C. Chan 16/02/2014
REUTERS
Gun deaths in Florida
Number of murders committed using firearms
873
800
600
400
200
0
721
2005
Florida enacted its 'Stand Your Ground' law
1990s
2000s
2010s
Source: Florida Department of Law Enforcement

75

Can you be a “good” visualization designer in a “bad” (political) world?

76

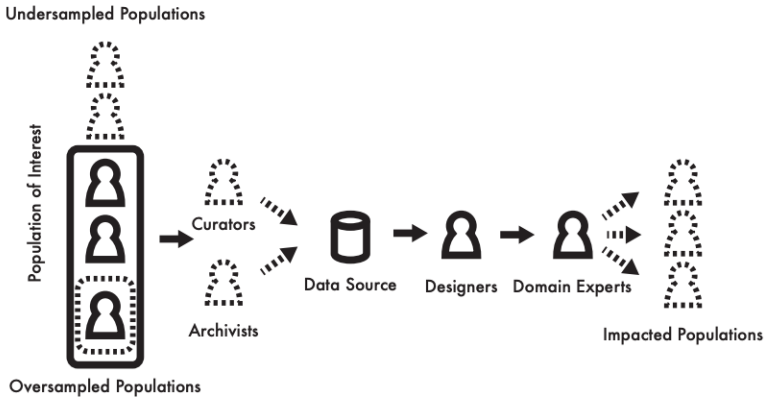
Well, yes...

- But it doesn’t mean we don’t get criticized
- Also doesn’t mean we don’t deserve to be criticized
- We don’t always recognize the bias in ourselves

77

Ethical Dimensions of Visualization Research

Michael Correll
Tableau Research
Seattle, WA
mcorrell@tableau.com



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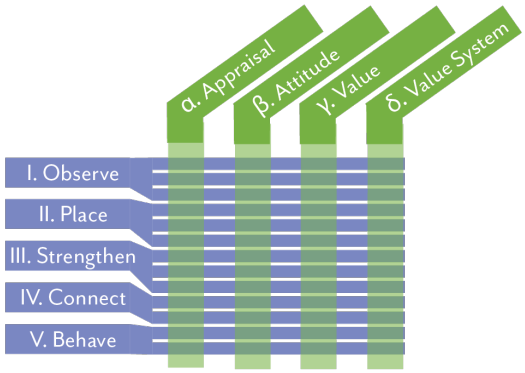
I. Observe
(i) Perceive
(ii) Consider
(iii) Attend to

II. Place
(i) Agree
(ii) Accept
(iii) Prefer

III. Strengthen
(i) Support
(ii) Believe
(iii) Commit to

IV. Connect
(i) Compare
(ii) Rank
(iii) Rate

V. Behave
(i) Demonstrate
(ii) Practice



α. Appraisal
A value judgement of a data fact

β. Attitude
A subjective disposition towards something

γ. Value
Deeply held belief

δ. Value System
Set of beliefs

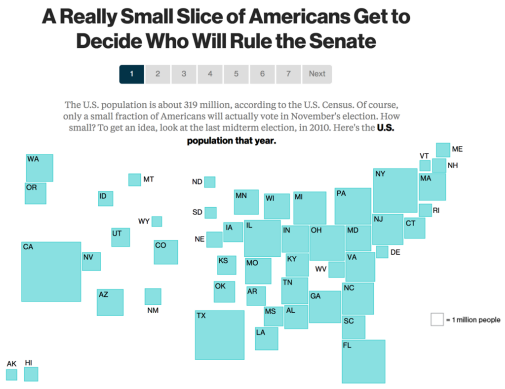
"Affective Learning Objectives for Communicative Visualizations" Lee-Robbins & Adar, 2022.

79

Takeaways

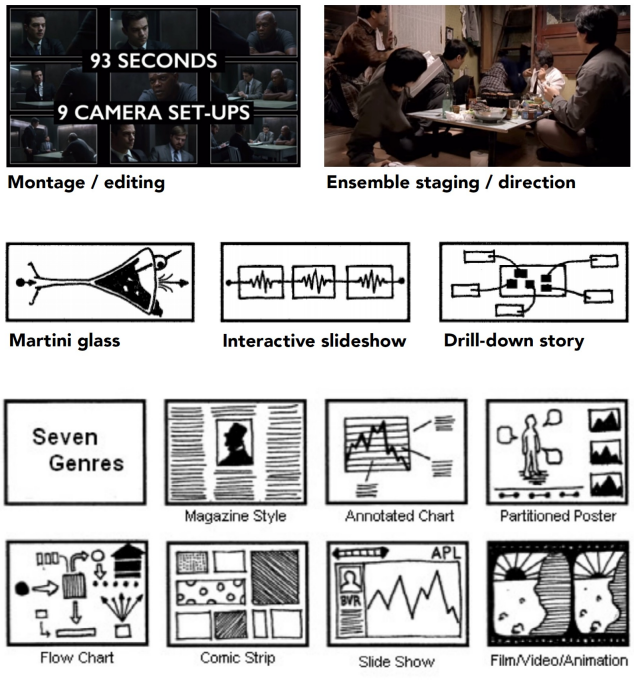
- Communicative visualization can be viewed like learning
- We can leverage ideas from storytelling, narrative, emphasis, structure, etc.
- Communicative visualizations are not complex
  - Invariably political whether we want them to be or not

80

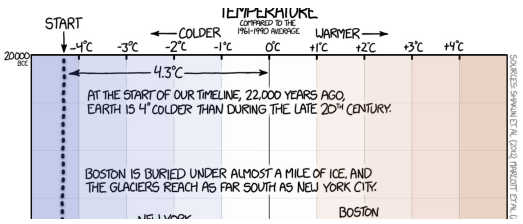


https://tinyurl.com/yaehpraw

What techniques are they using?
How are they directing attention?
What are the likely objectives?
Does it work?

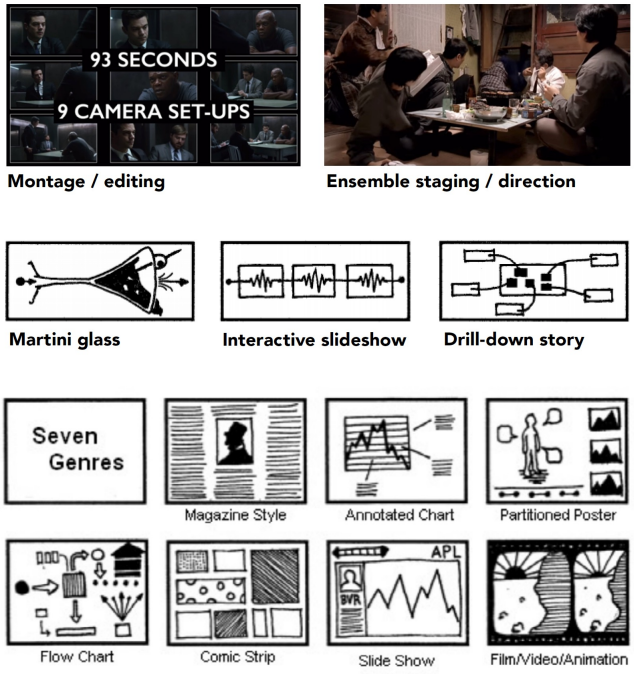


81

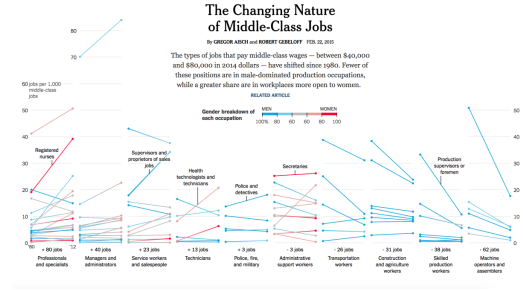


https://xkcd.com/1732/

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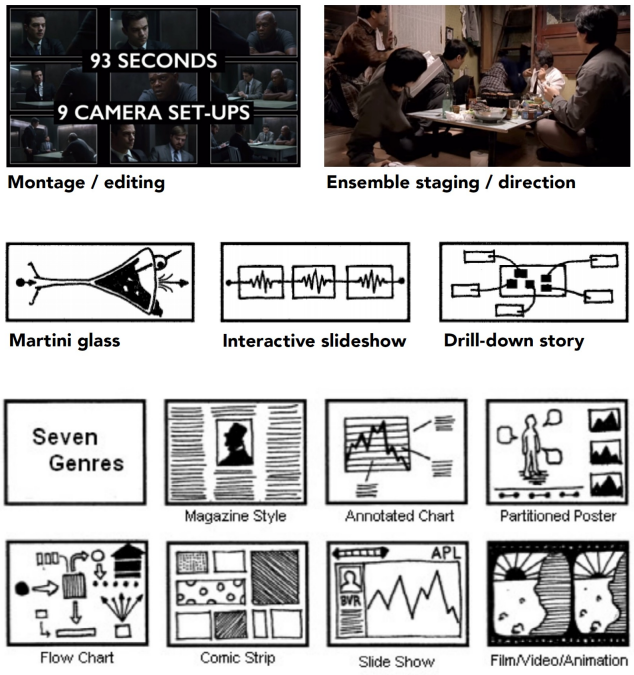


82

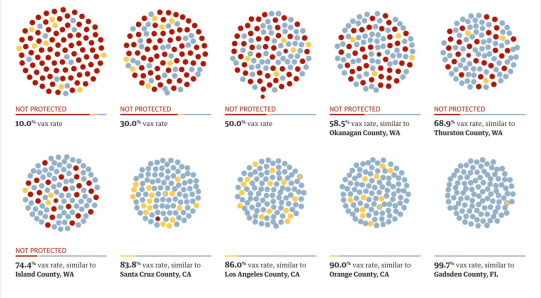


https://nyti.ms/2k5jxOf

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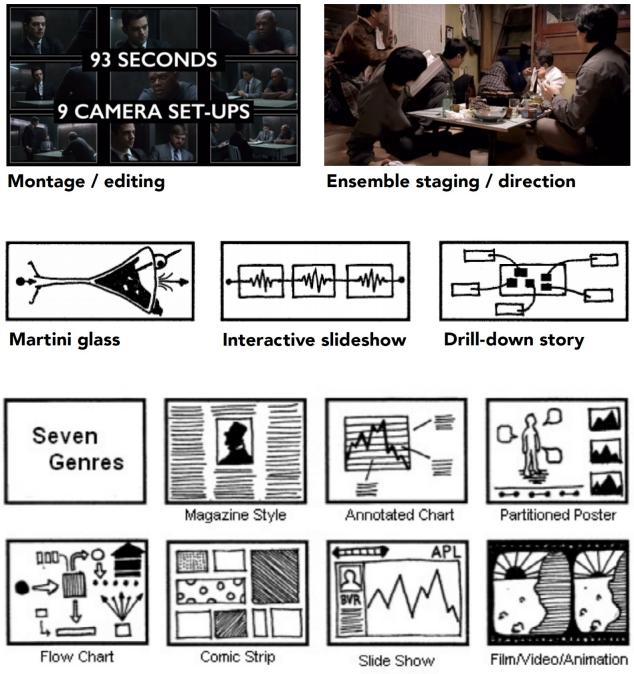


83

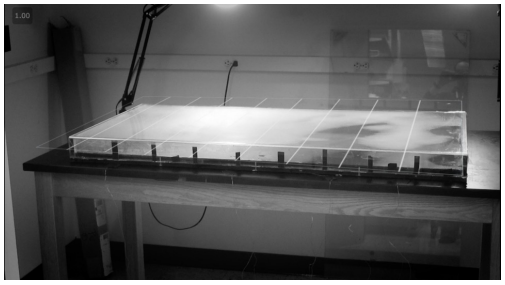


https://tinyurl.com/mjd5sv9

What techniques are they using?
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What are the likely objectives?
Does it work?

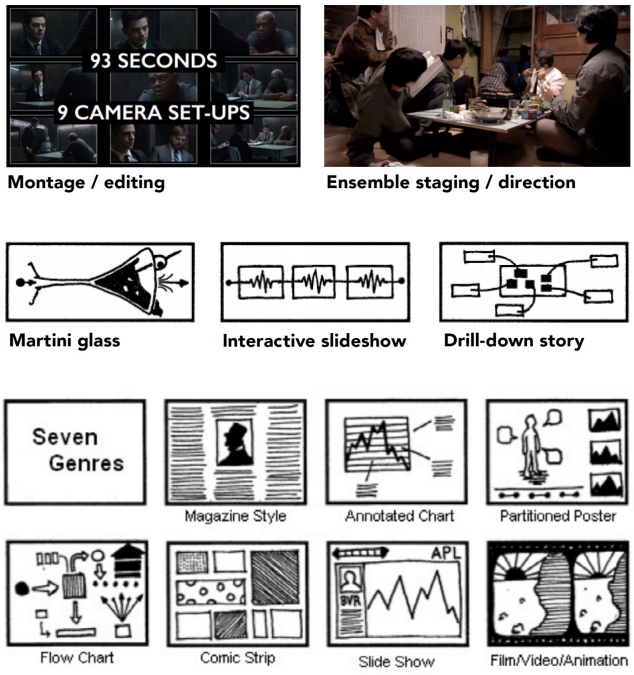


84



https://vimeo.com/180908160

What techniques are they using?
How are they directing attention?
What are the likely objectives?
Does it work?



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